

Preparation of novel marine antifouling polyurethane coating materials

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Abstract Polyurethane coating materials with different compositions of low surface energy polydimethylsiloxane and degradable poly(L-lactic acid) were synthesized by three major steps. Initially, the hydroxylation-terminated poly(L-lactide)-functionalized graphene (G-g-PLLA) was prepared by ring-opening polymerization of L-lactic acid using the phenol-functionalized graphene (G-f-OH), which was prepared by 1,3-dipolar cycloaddition reaction of graphene and 3,4-dihydroxybenzaldehyde when *N*-methylglycine and tin octoate were used as initiator and catalyst, respectively. Subsequently, isocyanate-terminated polyurethane prepolymer with polydimethylsiloxane was obtained by condensation polymerization of polydimethylsiloxane and isocyanate-terminated polyurethane prepolymer that was obtained by the condensation polymerization of 4,4'-diphenylmethane diisocyanate and 1,4-butane diol. Finally, the novel polyurethane coating materials were prepared by the condensation polymerization of G-g-PLLA and isocyanate-terminated polyurethane prepolymer with polydimethylsiloxane. These synthesized materials

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were carefully analyzed with Fourier transform infrared spectroscopy (FTIR), proton nuclear magnetic resonance spectra (^1H NMR), field-emission scanning electron microscopy (SEM), and high-resolution transmission (TEM). In addition, the water contact angles were measured. It was found that the surface free energy of the polyurethane coating materials decreased from 52.19 to 11.74 N/m² with the increase of polydimethylsiloxane content from 0 to 20% and the water contact angle of the polyurethane coating materials increased from 71° to 108°. Moreover, the mechanical property was investigated. The studies also demonstrated that functionalized polyurethane was able to hydrolyze in seawater and the hydrolysis rate decreased as the PDMS content increased. At the same time, simulative ocean hanging plate experiment confirmed that the novel polyurethane coating materials exhibited a good antifouling performance, which indicated that the functionalized polyurethane has a potentiality in marine antifouling coating application.

Keywords Marine antifouling · Low surface energy · Biodegradation · Polyurethane

Introduction

Biofouling, which is defined as the colonization of a diverse set of microorganisms, plants, algae, and animals on surface immersed in seawater [1], causes serious practical and economical consequences. It can increase the resistance of ship navigation, accelerate the corrosion of the hull due to the coatings on the metal surface are destroyed, block the cooling cycle pipe, increase the number of repairing hull, which resulting in a large amount of time, and resources are consumed. Thus, studies on antifouling coatings have been done. Tributyltin self-polishing coatings were used to prevent biofouling for protection of ship hulls in the past. However, tributyltin compounds have shown deleterious effects on the environment [2]. Tin-based antifouling coatings have been developed for antifouling, which is not an extraordinarily efficient biocide. Some ways to prevent fouling are to add antifoulants into polymer chain; however, its antifouling ability needs to be improved since only when their concentration is sufficient they can exhibit antifouling ability [1]. Although the incorporation of a nature's antifouling extract directly into a coating can exhibit antifouling ability, antifouling products from marine organisms are limited by insufficient supply [3]. Other strategies include mimic natural antifouling surfaces [4], and the superhydrophobic property of the lotus leaf. However, current fabrication methods are expensive and the application methods are complicated [5]. Therefore, an increasing level of attention has been given to the development of low surface energy and degradable coating with tailored antifouling property.

Environment-friendly poly(L-lactide) (PLLA) derived from renewable sources has obtained extensive attention as a new class of materials in the fields of biomedicine and environmental protection, such as biomedical devices, agricultural mulch films [6], packaging [7], and marine antifouling coating [8] due to its good biocompatibility and biodegradability [9]. There are several ways to synthesize

PLLA. The two most common methods are ring-opening transesterification polymerization and dehydration condensation. However, the dehydration condensation is difficult to prepare high-molecular-weight polymers and the polymerization reaction is difficult to control as the water is continuously generated during the preparation process, while ring-opening polymerization has obvious advantages. There is no water in the reaction process and the molecular weight and its distribution can be controlled. Samarajeewa et al. [10] fabricated P(NAS-*co*-NAM)-*b*-PLLA copolymer materials, which the PLLA was obtained by ring-opening polymerization in the copolymer materials. They demonstrated that this material has excellent degradation properties and the enzyme-triggered selective excavation was also demonstrated. To broaden the application of PLA, many attempts have been focused on the incorporation of low amounts of nanoparticles into the PLA matrix. Among nanoparticles, graphene is an interesting option. Graphene, a honeycomb lattice material in which planar carbon atoms are closely bounds, has shown significant application prospects in the field of chemistry, physics, materials, and electronics [11] due to its outstanding electrical, optical, thermal, and mechanical properties [12]. The incorporation of graphene has resulted in the improvement of thermal, mechanical, and electrical properties of polymer matrices [13]. In addition, graphene and graphene-based nanocomposites have also been used in bacteria detection and antibacterial applications [12].

Polydimethylsiloxane (PDMS) is known to possess high flexibility, low cost, biocompatibility [14], low surface energy [15], and good thermal stability [16], etc., which provides an opportunity to further develop their applications. The physical nature of PDMS materials enables PDMS coatings to remove biofouling easily when biofouling does occur. It is generally believed that the good fouling release performance of PDMS coatings is due to its low surface energy and low elastic modulus [17]. On the other hand, the PDMS coating can also achieve the effect of drag reduction due to its low surface energy. Kim et al. [18] used pore-type poly(dimethylsiloxane) microstructures to reduce the contact area between water and the tube wall, and illustrated that a tube with a higher contact angle exhibits relatively lower skin friction and delays the velocity saturation point better than a tube with a lower contact angle in the laminar flow region. Gao et al. [19], using a depressurization-driven method to form a bubble-covered surface on the wall of PDMS microchannel, found that the flow boundary condition at the bubble-covered surface changes from no slip to slip and causes a dramatic drag reduction. Therefore, the PDMS coatings are being developed as a marine antifouling coating. Antifouling coatings have been established to avoid the accumulation of biofouling organisms on ship hulls and on cement slab and wood construction in the aqueous environment.

Polyurethane are a class of thermoplastic elastomers, which its structure–property relationships have drawn much attention [20–23]. Due to their excellent mechanical properties, biocompatibility and polyurethanes (PUs) have been extensively used in various fields as adhesives, coatings, foams, and packaging materials [24, 25]. Particularly, they show excellent protein adsorption resistance. Furthermore, combining polyurethane with low surface energy and biodegradable polymers is expected to obtain outstanding antifouling ability and excellent

mechanical properties and adhesion [26], which have high potential applications for marine antifouling coatings.

In this contribution, the G-g-PLLA with controlled degradation property was incorporated into antifouling polyurethane coating based on PDMS to obtain improved antifouling performance coating. The objective of this study is to synthesize polyurethane coating materials with low surface energy and degradable properties, and its low surface energy, water contact angle, mechanical property, and antifouling performance were also investigated. In addition, the degradation of the polyurethane coating materials can be controlled by regulating the content of graphene, as the graphene have an effect on the crystallization of the poly(L-lactic acid), which resulting in the controlled degradation property of the polyurethane coating materials can be achieved [25]. The marine antifouling polyurethane coating material described here is a promising candidate for a powerful protocol for marine antifouling, because it has the following advantages: (1) the production is environment-friendly and non-toxic, (2) the antifouling coating possess both low surface energy and degradable properties, resulting in the antifouling ability of coating was better, (3) the synthetic method is simple, and its cost is low, not using expensive materials, and (4) this improvement on the antifouling performance of the coating based on low surface energy and degradation might lead to a new approach to further develop cost-effective and highly efficient coating for marine antifouling.

Experimental

Materials

Natural flake graphite with an average particle size of 300 mesh and a purity of > 99% was supplied by Qingdao Haida Graphite Co, Ltd, China. Dimethylformamide (DMF, 99%, Shanghai Chemical Industrial Co, Ltd, China) was purified by stirring over KOH for 48 h and then distilled under reduced pressure. L-Lactide (LLA), obtained from Aldrich, was recrystallized twice from anhydrous ethyl acetate and stored under dry nitrogen prior to use. 1,4-Butane diol (1,4-BDO) from Aldrich was dried under reduced pressure for 2 h prior to use. Polydimethylsiloxane were used as received from ABCR. Toluene, obtained from Aldrich, mixed with small pieces of sodium metal, zeolite, and indicator diphenyl ketone, was relaxed until the solution turns blue, and then distilled under atmospheric pressure prior to use. 3,4-Dihydroxybenzaldehyde, *N*-methylglycine, 4,4'-diphenylmethane diisocyanate (MDI), and stannous 2-ethylhexanoate ($\text{Sn}(\text{oct})_2$) were purchased from Aldrich and used without further purification. All other reagents and solvents were purchased from commercial suppliers and used as received.

Physical measurements

Fourier transform infrared (FTIR) spectra were measured with a PE Spectrum One B instrument, and the spectra were collected at 64 scans with a spectral resolution of 4 cm^{-1} by KBr disk method. Raman spectra were collected on a Renishaw inVia

plus laser Raman spectrometer with an excitation laser wavelength of 532 nm (5 mW) at room temperature. Nuclear magnetic resonance (NMR) spectra were obtained through the equipment of NMK/MHz. SEM measurements were carried out on a Nova NanoSEM 230 field-emission scanning electron microanalyzer. TEM measurements were conducted on a JEM-2100F TEM microscope. The contact angle and the surface free energy were measured by the sessile-drop method using a Kruss drop shape analysis system, DSA 100 (Kruss GmbH, Hamburg, Germany). The whole measuring process was monitored by an imaging apparatus. Tensile strength test was carried out on a tensile tester (Model DAGE 4000, Britain) at room temperature with a speed of 50 mm min^{-1} . All measurements have an average of three runs. For the static hydrolysis experiment, a solution of 20% (w/v) polyurethane in refined tetrahydrofuran was dripped onto an epoxy resin panel ($25 \text{ mm} \times 25 \text{ mm}$ in size). The panels coated with G-g-PLLA-*b*-PDMSs-PU were kept at room temperature for 5 days to completely volatilize the THF. The weight (m_o) of each dried coating together with its panel was then measured before dipping into natural seawater, which was renewed once every 2 weeks. After 7 days, the panel was taken out, rinsed with deionized water, and dried at 60°C in a vacuum oven for 24 h, and the weight (m_s) of the panel was measured again. The mass loss was designated as ($m_s - m_o$). The dispersion experiment: taking 50 mg of graphene oxide, 50 mg of graphene, and 50 mg of graphene-f-OH added to 10 mL of water, 10 mL of DMF and 10 mL of DMF, and ultrasonic treatment at room temperature for 30 min. Simulative ocean hanging plate experiment: we used the natural seawater from the inner Xiamen bay in China and it was dumped into a pool ($1 \text{ m} \times 1.5 \text{ m} \times 1.5 \text{ m}$). Different types of resins were dissolved in tetrahydrofuran and then were applied to the epoxy plate with thickness of 0.5 mm; it was put into the seawater to soak together at room temperature for 60 days after it was dried.

Preparation of graphite oxide and graphene

Graphite to graphite oxide: graphite oxide, a pseudo-two-dimensional solid in bulk form, was prepared from natural graphite powder by oxidation with KMnO_4 in concentrated H_2SO_4 according to Hummers' method. Concentrated H_2SO_4 (46 mL) was poured into the 250 mL three-neck flask and stirred in an ice bath until the temperature dropped to $0\text{--}3^\circ\text{C}$. Natural graphite (2.0 g) and NaNO_3 (1.0 g) were added and stirred uniformly. KMnO_4 (6.0 g) was gradually added with stirring and cooling to keep the temperature below 20°C . The solution was heated to $35 \pm 3^\circ\text{C}$ and maintained for 30 min. Then, distilled water was slowly added, and the temperature was kept lower than 100°C . After 15 min, this reaction was terminated by addition of a large amount of distilled water and 30% H_2O_2 solution (5 mL). The mixture was filtered and washed with 5% HCl aqueous solution and water. The sample of graphite oxide was obtained after drying. Graphite oxide to graphene oxide: 100 mg graphite oxide was dispersed in 100 mL of water to create a yellow–brown dispersion and the exfoliation of graphite oxide to graphene oxide was achieved by sonication with a cylindrical tip for 30 min. Graphene oxide to graphene: the resulting homogeneous solution was mixed with 80 mL of hydrazine solution as a reducing agent, and 350 mL of ammonia solution was added to adjust

pH to around 10. After stirring for a few minutes, the mixed solution was heated at about 95 °C for 1 h. The reduced product was isolated by filtration through polytetrafluoroethylene membrane (0.22 mm), washed with water and methanol, and dried for 24 h in vacuum oven. Then, graphene, that is, the reduced graphene oxide, was obtained.

Functionalization of phenol-functionalized graphene (graphene-f-OH)

Graphene to graphene hydroxide: 20 mg of graphene, 200 mg of 3,4-dihydroxybenzaldehyde, and 200 mg of *N*-methylglycine were suspended in 50 mL of DMF and heated at 120 °C for 5 days. The mixture was then filtered through Millipore filters (0.45 mm) and washed thoroughly with DMF. The filtrate was sonicated in DMF for 1 h and then removed; the resulting suspension was filtered again through Millipore filters (0.45 mm). This procedure was repeated three additional times with sonication in (1) DMF, (2) 1:1 (v/v) C₂H₅OH/CHCl₃, and (3) diethyl ether. The remaining black solid was dried under vacuum (10^{−2} bar) for 3 days.

Synthesis of poly(L-lactic acid) grafted graphene polymer materials (G-g-PLLA)

Poly(L-lactic acid) grafted graphene macroinitiators were synthesized as shown in Fig. 1, using a procedure similar to those described previously for ring-opening polymerization of LLA to give PLLA–OH. The polymerization took place in

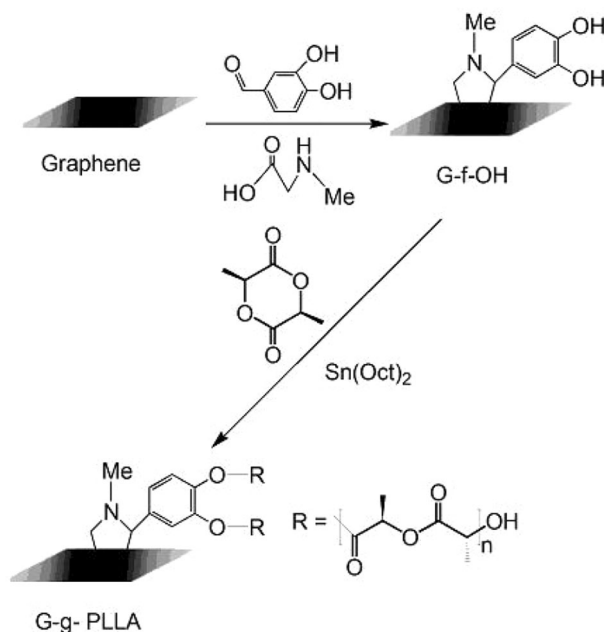


Fig. 1 Synthesis of graphene-grafted PLLA (G-g-PLLA)

anhydrous toluene solution at 120 °C using graphene-f-OH as initiator and tin octoate as catalyst. Specifically, 2 g of LLA were charged into the prepared flask and three vacuum-nitrogen cycles were carried out. Calculated amounts of freshly prepared solutions of tin octoate (1 mL) and graphene-f-OH (0.04 g), followed by 50 mL of dry toluene, were then added through a septum via a syringe under nitrogen flow. Finally, the system was closed under positive nitrogen pressure, and polymerization proceeded under stirring at 120 °C for 24 h. After reaction, the mixture was stopped by cooling to room temperature. The solidified reaction mixture was dispersed in 30 mL of methanol and then poured into 80 mL of methanol containing 1 mL of concentrated HCl to remove the Sn residue. The black solid was collected by filtration, washed with methanol, dried, redissolved in 10 mL of dry dichloromethane or chloroform, and precipitated dropwise into 100 mL of methanol containing 1 mL of concentrated HCl. The precipitate was collected by filtration, washed with methanol, and dried in vacuo at 30 °C for 72 h.

Synthesis of isocyanate-terminated polyurethane prepolymer with polydimethylsiloxane (PDMS-*b*-PU)

The isocyanate-terminated polyurethane prepolymer with polydimethylsiloxane was synthesized via a condensation reaction in anhydrous toluene solution under a nitrogen atmosphere, as shown in Fig. 2. First, the 1,4-BDO (0.011 mol) and the 4,4'-MDI (0.015 mol) were added into a four-neck flask to equipped with a mechanical stirrer, a thermometer, a reflux condenser, and using anhydrous toluene

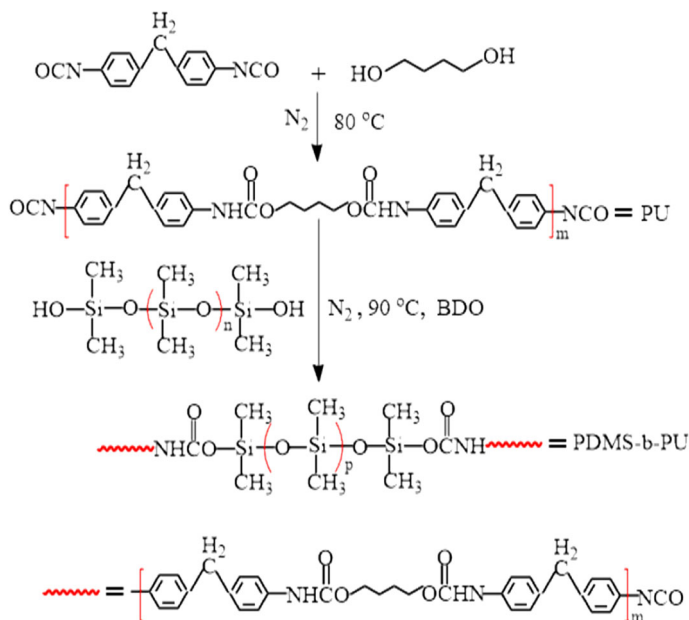


Fig. 2 Synthesis of isocyanate-terminated polyurethane prepolymer with polydimethylsiloxane (PDMS-*b*-PU)

as solvent. The reaction was carried out at 80 °C for 2 h under N₂ atmosphere with agitation. After adding the hydroxyl polydimethylsiloxane, the reaction was continued at 90 °C for another 6 h. Crude production of isocyanate-terminated polyurethane prepolymer with polydimethylsiloxane was prepared after cooling the reaction system to ambient temperature; then, the crude production was added into its eight times volume hexane to precipitate the crude production. Subsequently, the production was dried at 50 °C in vacuum drying box. On the basis of the method, a series of isocyanate-terminated polyurethane prepolymer with polydimethylsiloxane were prepared.

Synthesis of the novel polyurethane copolymer (G-g-PLLA-PDMS-*b*-PU)

The synthesis of G-g-PLLA-PDMS-*b*-PU is shown in Fig. 3. 50 mL anhydrous toluene and a certain amount synthesized G-g-PLLA were first added into a four-neck flask equipped with a mechanical stirrer, a thermometer, a reflux condenser, and a nitrogen catheter. The end-capped polyurethane with isocyanate group prepolymer was added into the system after the prepolymer was dissolved completely and then the reaction was carried out at 80 °C for 2 h under N₂ atmosphere with agitation. Crude production was prepared after cooling the reaction system to ambient temperature. After the crude production was filtered, it was dried at 50 °C for 12 h in vacuum, and then, the novel polyurethane copolymer was obtained.

Results and discussion

Preparation of the novel polyurethane copolymer (G-g-PLLA-PDMS-*b*-PU)

Our strategy to synthesize novel polyurethane copolymers (G-*g*-PLLA-PDMS-*b*-PU) consisted of three major steps, and they are shown in Figs. 1, 2 and 3, respectively.

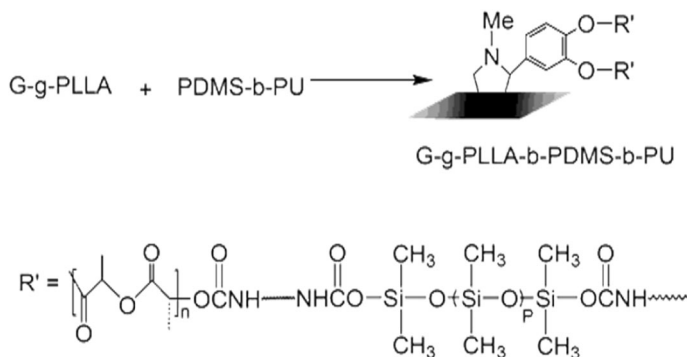


Fig. 3 Synthesis of the novel hybridization polyurethane (G-g-PLLA-PDMS-*b*-PU)

Structure analysis

The resulting phenol-functionalized graphene (graphene-f-OH) was characterized by FTIR. Figure 4 shows the FTIR spectra of pristine graphene and phenol-functionalized graphene (graphene-f-OH). As illustrated in Fig. 4, the FTIR spectrum of graphene-f-OH shows C–H stretch features at 2950, 2922, and 2852 cm^{-1} that do not appear in the spectrum of pristine graphene (curve a in Fig. 4). In addition, apparent C–C stretch bands and a weak aromatic C–H band were observed at 1100–1700, 61, and 3040 cm^{-1} , respectively. The C–O stretch band was observed at 1120 cm^{-1} and the O–H stretch band was observed at 3600–3700 cm^{-1} , both of which are characteristic of phenols. The FTIR spectral analysis provides support for successful functionalization as procedures illustrated in Fig. 1 by 1,3-dipolar cycloaddition reaction using 3,4-dihydroxybenzaldehyde.

Figure 5 illustrates SEM image of G-f-OH. As shown in Fig. 5, it can be found that the graphene-f-OH shows a flake layer structure throughout the scanned areas, which indicates that the compatibility between the graphene and organic solvents was significantly improved. The result of the SEM image of G-f-OH also confirms that the G-f-OH have been successfully synthesized via 1,3-dipolar cycloaddition reaction using 3,4-dihydroxybenzaldehyde.

Figure 6 presents the Raman spectra of the synthesized graphene and graphene-f-OH. It shows two characteristic peaks with the D band (1347 cm^{-1}) and the G band (1603 cm^{-1}). Usually, the intensity ratio (I_D/I_G) is usually used to evaluate the integrity of the graphene. Their intensity ratios (I_D/I_G) in graphene and graphene-f-OH are 1.02 and 0.75, respectively. That means that the integrity of the graphene in graphene-f-OH which graphene was covalently functionalized with phenol groups by a 1,3-dipolar cycloaddition reaction [27] was decreased.

In addition, the dispersion experiment was carried out to evaluate the functionalization effect. The photographs of graphene oxide, graphene, and

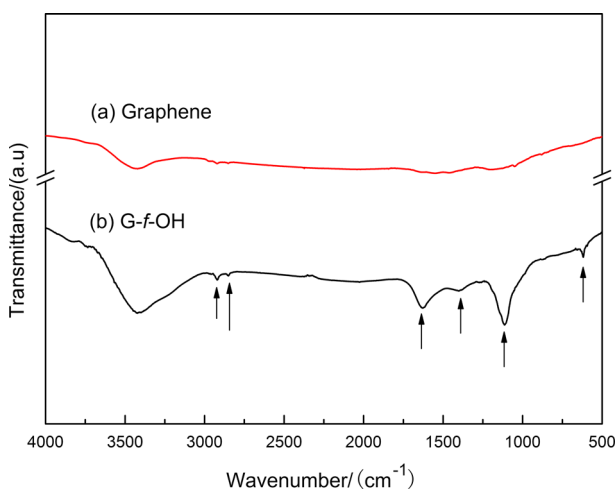


Fig. 4 FTIR spectrum of (a) graphene and (b) graphene-f-OH (G-f-OH)

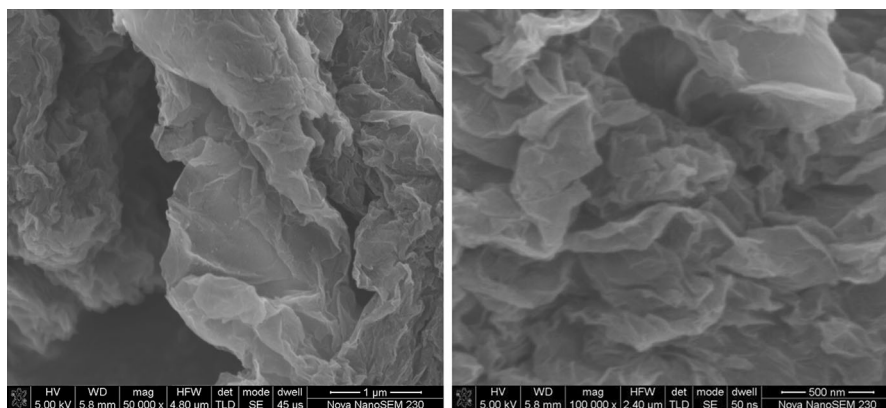


Fig. 5 SEM image of the pristine graphene and G-f-OH

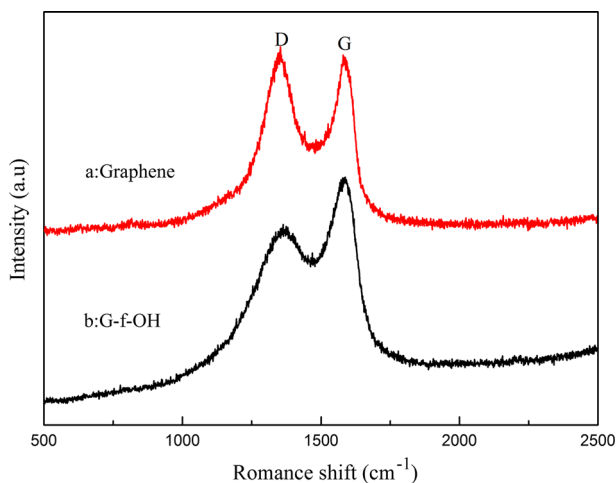


Fig. 6 Raman spectra of the graphene and G-f-OH

graphene-f-OH are illustrated in Fig. 7, which were taken after the related graphene oxide dispersed in water, graphene, and graphene-f-OH samples dispersed in DMF a month later. In contrast, there are some additional carbonyl and carboxyl groups located at the edge of the graphene oxides, which make graphene oxides strongly hydrophilic, allowing them to readily disperse in water. However, there are few additional carbonyl and carboxyl groups located at the edge of the graphene, which is incompatible with most organic solvents. It can be obviously observed from Fig. 7b that the pristine graphene had settled in the test tube. The introduction of phenolic substituents makes the graphene more compatible with other solvents, rendering them dispersible in DMF (Fig. 7c). These results also show the successful functionalization of graphene with phenolic groups, indicating that the reaction had taken place and afforded the desired graphene-f-OH.

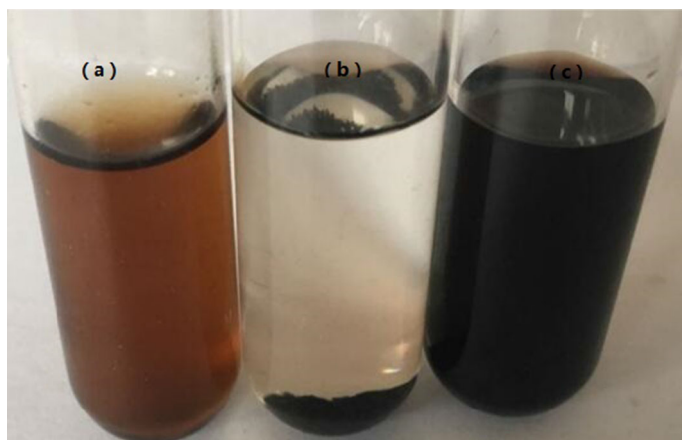


Fig. 7 Photographs of (a) graphene oxide in water, (b) graphene, and (c) graphene-f-OH in DMF

The FTIR spectra of phenol-functionalized graphene (graphene-f-OH) and PLLA-grafted graphene polymer materials (G-g-PLLA) shown in Fig. 8. The appearance of peaks at 1040 and 3440 cm^{-1} can be assigned to the stretching vibrations of C–O and –OH, respectively, which appear in the spectrum of PLLA/g-graphene. The new band at 1440 cm^{-1} is attributed to the bending vibration of –CH₃, and the new band at 1670 cm^{-1} is associated with the coupling of C–O–C and –CH– vibration modes. These results clearly confirmed that the L-lactide acid molecules were covalently linked to graphene, suggesting successful surface functionalization of the graphene with an OH-terminated group.

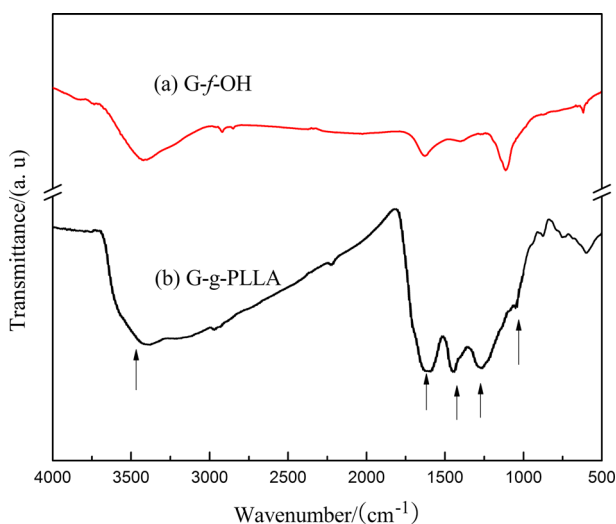


Fig. 8 FTIR spectrum of (a) G-g-OH and (b) G-g-PLLA

^1H NMR spectroscopy was used to further testify the successful graft of PLAA into graphene. As shown in Fig. 9, Peak a, e, and f located at 1.28, 4.35, and 1.36 ppm were contributed by the proton of $-\text{CH}-$, $-\text{CH}_3$, and $-\text{CH}_2-$ connected with nitrogen, respectively. Peak b and d located at 5.19–5.23 and 1.53–1.62 ppm were produced by the proton of $-\text{CH}-$ and $-\text{CH}_3$ in the polymer chain, respectively, and peak c located at 3.73 ppm is attributed to the proton of terminal $-\text{OH}$. Peak i, g, and h were located at 7.05, 7.26, and 7.54 ppm and were assigned to the protons on phenyl groups. The ratio of the area of Peak b to that of Peak e is approximately 35, which indicates that the grafting length is approximately 105 repeat structure units. The ^1H NMR results confirmed that the G-g-PLLA composites have been successfully synthesized via ring-opening polymerization.

The morphology of G-g-PLLA was also investigated by TEM, as shown in Fig. 10. Graphene-f-OH shows a flake layer structure, which is consistent with the results of SEM of the G-g-OH. It can be found that the surface on the graphene-f-OH attached with the slice polymer uniformly, not occurring the phenomenon of the conglomeration. It is postulated that these structures correspond to graphene-attached PLLA.

The ^1H NMR spectra of PDMS-*b*-PU are shown in Fig. 11. The peaks at 0.87 and 0.07 ppm correspond to the signals of $-\text{CH}_3$ in the PDMS. The $-\text{CH}_2-$ groups are located at 1.23–1.26 ppm. The $-\text{CH}_2-$ groups at site c, $-\text{OCONH}-$ protons, and aromatic protons were found at 1.43, 3.73, and 7.26 ppm, respectively. The ^1H NMR results confirmed that the PDMS-*b*-PU have been successfully synthesized via condensation polymerization.

The FTIR spectra of PDMS-*b*-PU are shown in Fig. 12a. As illustrated in Fig. 12a, the FTIR spectrum of PDMS-*b*-PU shows $-\text{NH}-$ stretching mode at 3320 cm^{-1} . In addition, apparent $-\text{NCO}$ stretch bands and the stretching vibration of carbonyl $\text{C}=\text{O}$ in $-\text{NH}-\text{COO}-$ were observed at 2314 and 1703 cm^{-1} ,

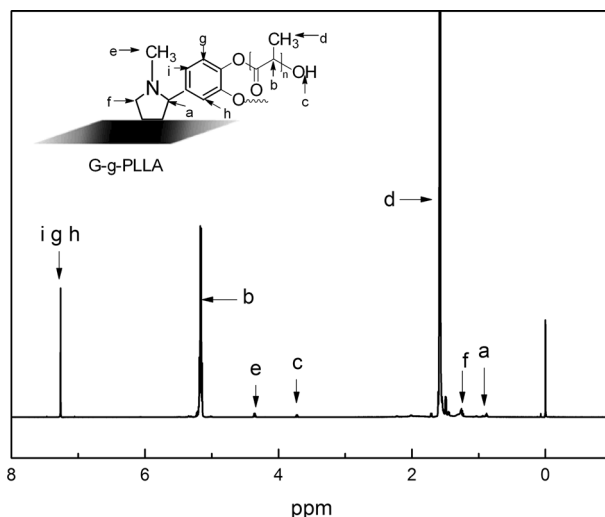


Fig. 9 ^1H NMR spectrum of G-g-PLLA in CDCl_3

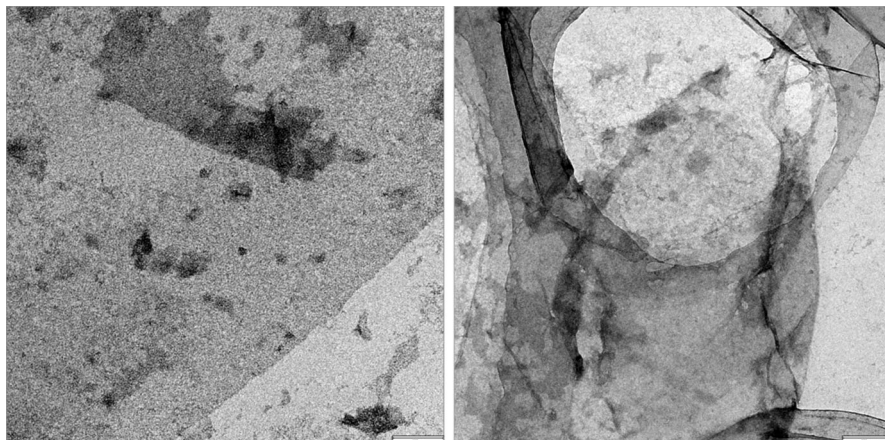


Fig. 10 TEM images of the G-g-PLLA

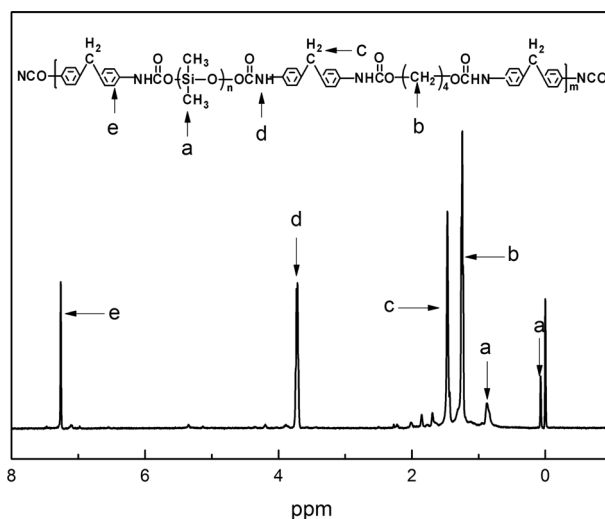


Fig. 11 ^1H NMR spectrum of the PDMS-*b*-PU in CDCl_3

respectively. The new band at 1540 cm^{-1} was attributed to the characteristic absorption peak of $-\text{NHCOO}-$. The Si–O stretch band at 1058 cm^{-1} and the Si–C bending vibration at 1219 cm^{-1} , both of which are characteristics of siloxane. All these results clearly confirm that the PDMS-*b*-PU was successfully prepared.

The FTIR spectra of the novel polyurethane copolymer (G-g-PLLA-PDMS-*b*-PU) are shown in Fig. 12b. As illustrated in Fig. 12b, the band at 3320 cm^{-1} is attributable to the N–H stretching mode and the peaks at 820 cm^{-1} can be assigned to the bending vibration of aromatic C–H. The band at 1700 cm^{-1} is attributable to the stretching vibration of the carbonyl C=O in PLLA, which indicates the successful chemical reaction between the OH-terminated poly(L-lactide)-

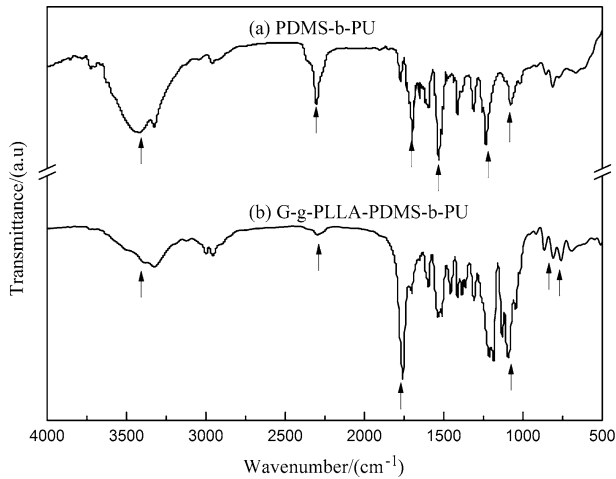


Fig. 12 FTIR spectra of (a) PDMS-*b*-PU and (b) G-*g*-PLLA-PDMS-*b*-PU

functionalized graphene and the NCO-terminated polyurethane prepolymer and the peaks at 819 and 858 cm^{-1} can be assigned to the C–Si–C stretching vibration. In addition, the band at 1080 cm^{-1} is associated with the Si–O–Si stretching mode in PDMS. All these results clearly confirmed that the novel degradable and low surface energy polyurethane hybrid composite was successfully prepared via a combination of ring-opening polymerization and condensation polymerization.

Thermogravimetric analysis

Figure 13 illustrates the TG thermograms of G-*g*-OH, G-*g*-PLLA, and G-*g*-PLLA-*b*-PDMS-*b*-PU. It was observed from Fig. 13 that a little mass lost around 100°C

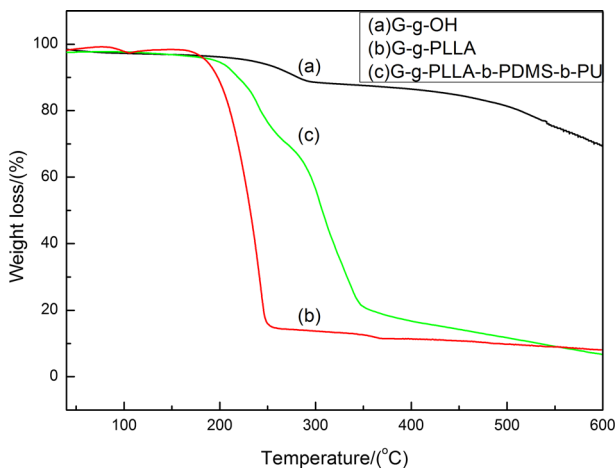


Fig. 13 TG thermograms of (a) G-*f*-OH, (b) G-*g*-PLLA, and (c) G-*g*-PLLA-*b*-PDMS-*b*-PU

due to the deintercalation of H₂O and solvent. The obtained G-OH sample shows a mass loss of approximately 25.3%, which is attributed to the decomposition of the grafted hydroxyl groups. Figure 13b also shows that G-g-PLLA degraded in one step from 180 °C, and a mass loss of 83.9% occurred from 170 to 300 °C, which is attributed to the decomposition of the PLLA. It can be seen from Fig. 13c that the G-g-PLLA-*b*-PDMS-*b*-PU degraded in two steps. The first is from 180 to 260 °C and the second is from 260 to 360 °C. There is a mass loss of 90.9% from 45 to 600 °C, which is reasonably attributed to the decomposition of the PLLA and polyurethane. Compared with the G-g-PLLA, a mass loss of 56% can be observed for the G-g-PLLA-PDMS-*b*-PU copolymer composite from 40 to 600 °C.

Water contact angle and low surface energy of G-g-PLLA-*b*-PDMS-PU film

The distilled water contact angle of coating samples was measured in this study and the surface energy of coating samples was calculated by Owens and Wendt method:

$$(1 + \cos \varphi) \gamma_L = 2\sqrt{\gamma_s^d \gamma_L^d} + 2\sqrt{\gamma_s^p \gamma_L^p}, \quad (1)$$

$$\gamma_s = \gamma_s^d + \gamma_s^p. \quad (2)$$

Here, φ is average of the distilled water contact angle of coating samples.

γ_L , γ_L^d , and γ_L^p are the distilled water surface free energy, the distilled water dispersion component, and the distilled water polar component, respectively.

γ_s , γ_s^d , and γ_s^p are the coating samples surface free energy, the coating samples dispersion component, and the coating samples polar component, respectively.

The water contact angle of polyurethane film with different polydimethylsiloxane (PDMS) content was shown in Fig. 14a. From this, it was found that the water

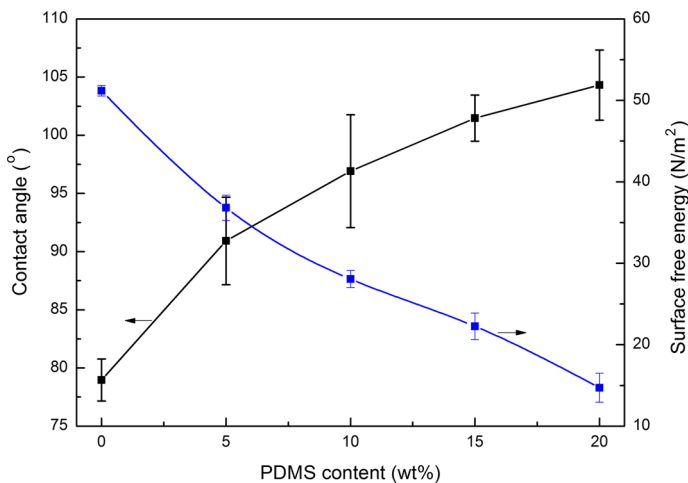


Fig. 14 Water contact angles and surface free energy change curves of the polyurethane film with different PDMS content

contact angle increased from 71° to 108° . It was also observed that the water contact angle increased linearly with the increase of polydimethylsiloxane (PDMS) content when polydimethylsiloxane (PDMS) content was below 10%. However, the contact angle increased at a slower rate when the polydimethylsiloxane (PDMS) content was higher than 10%. The trend of contact angle is inverse to the surface free energy. This phenomenon can be ascribed to that during the formation of polyurethane film, the migration quantity of polydimethylsiloxane-containing segments to the film surface work well for this polyurethane, because the movement of polydimethylsiloxane-containing segments with medium length chains is very easy, but when the PDMS content was higher than 10%, the migration quantity of polydimethylsiloxane-containing moieties to the film surface does not work effectively, because the movement of the polydimethylsiloxane-containing segments is restricted by the capacity of the film surface, and thus, water contact angle increases at a slower rate.

The surface free energy change curve of polyurethane film with different PDMS content was shown in Fig. 14b. It can be seen that the PDMS content influences surface free energy greatly; the curve descends sharply when its content is below 10%, but when the PDMS content is higher than 10%, the surface free energy decreases at a slower rate. It could be seen that the surface free energy decreases from 52.19 to 11.74 N/m² with the PDMS content from 0 to 20%, which reveals that this polyurethane can obtain good antifouling materials. This change in surface free energy trend indicated that the water resistance of the polyurethane has been improved greatly.

Mass loss of G-g-PLLA-*b*-PDMS-PU film

Figure 15 shows the mass loss of different PDMS-*b*-PU-*b*-G-PLLA in natural seawater at room temperature. As can be seen from the curve, compared with 20%

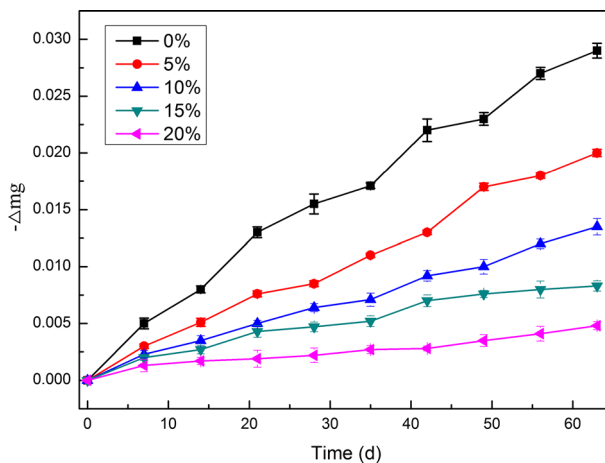


Fig. 15 Mass loss of the polyurethane film with different PDMS contents in natural seawater

PDMS-*b*-PU-*b*-G-*g*-PLLA, other PDMS-*b*-PU-*b*-G-*g*-PLLA panels generally lose weight faster, indicating a quicker hydrolytic degradation. Moreover, the hydrolytic degradation decreases with the increase of PDMS content. This phenomenon is ascribed to the hydrophobic effect of PDMS in the PU, resulting in decrease of hydrolysis ability of ester group in the PLLA main chain. Thus, we can design a surface with a controlled self-polishing rate by adjusting the content of PDMS.

Mechanical properties of G-*g*-PLLA-*b*-PDMS-PU film

The mechanical properties of G-*g*-PLLA-*b*-PDMS-*b*-PU film can be appraised from Fig. 16. The curves indicate the effects of PDMS content in the PUs on the tensile strength and extensibility, respectively. With increasing of PDMS content, the tensile strength decreased from 21 MPa for the sample 0% G-*g*-PLLA-*b*-PDMS-*b*-PU to 14 MPa for the sample 20% G-*g*-PLLA-*b*-PDMS-*b*-PU. It could be seen that the extensibility increases with increasing the PDMS content from 515% for the sample 0% G-*g*-PLLA-*b*-PDMS-*b*-PU to 658% for the sample 20% G-*g*-PLLA-*b*-PDMS-*b*-PU, which clearly showed that the extensibility of the film had been improved greatly due to the introduction of PDMS backbone. It could also be seen that the tensile strength and extensibility change at a faster rate when PDMS content was higher than 7.5%, indicating that PDMS has greater effect on the mechanical properties when PDMS content is above 7.5%.

Antifouling performance of G-*g*-PLLA-*b*-PDMS-PU film

Figure 17 shows the relationship between PDMS content and antifouling performance. All these polyurethane templates were soaked in seawater for 2 months. As can be seen, the blank template is the most serious pollution, which was covered

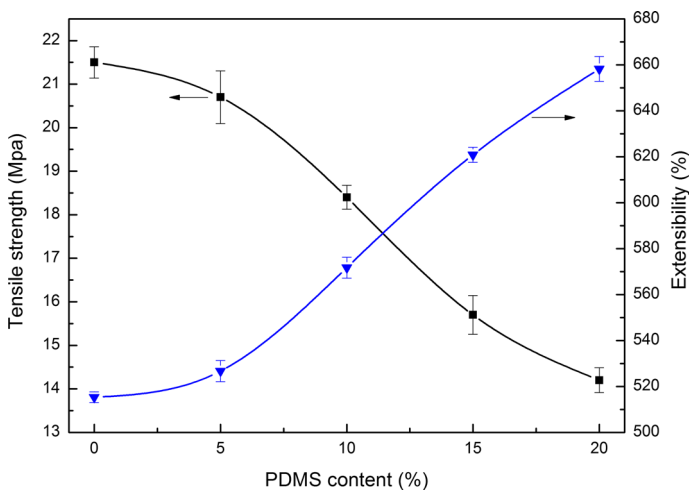


Fig. 16 Tensile strength and extensibility change curves of the polyurethane film with different PDMS contents

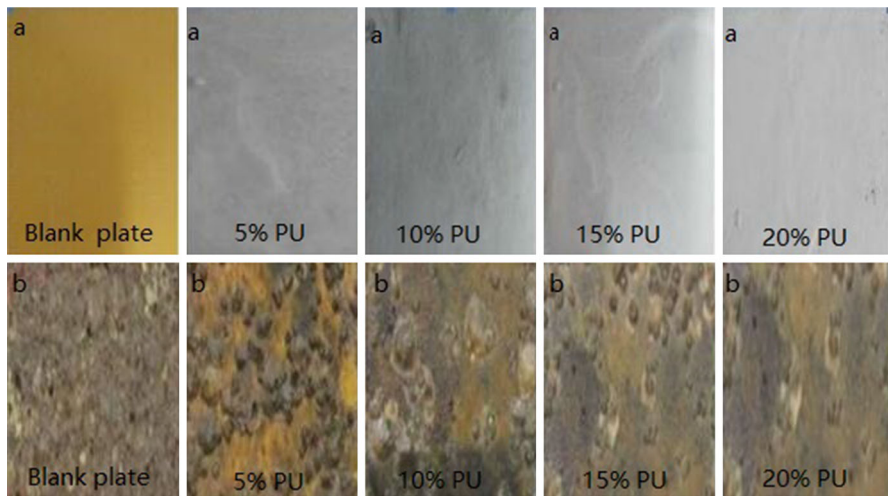


Fig. 17 Simulative ocean hanging plate experiment of **b** polyurethane templates with different PDMS contents for 60 days and **a** polyurethane templates with different PDMS contents for 0 day

with fouling on its surface. The number of fouling grown on the surface dramatically decreases as the PDMS content increases, indicating that the antibiofouling ability increases with the PDMS content. The reasons that polyurethane rein had a good antifouling effect were major two. First of all, the crystallinity of the material was reduced, and the degradation rate of the polyurethane was improved due to the graphene with antibacterial properties [28] and PLLA with degradable properties were both introduced into the polyurethane, which can lead to a self-renewal surface polishing the attached living organisms or inorganics away [9]. Second, the antifouling ability was improved with the increase of PDMS content of polyurethane, because the PDMS content influences the contact angle and surface free energy, and then, fouling adhesion force is reduced, resulting in the fouling is difficult to adhere or the fouling fell off under the effect of external force. In other words, the combination of degradable G-g-PLLA and hydrophobic PDMS in the main chain improved degradation and hydrophobicity, leading to the enhancement of antibiofouling.

Conclusions

In this study, a novel PLLA-grafted graphene was synthesized via ring-opening reaction. The synthetic method for the preparation of degradable and low surface energy polyurethane containing PLLA and polydimethylsiloxane was successfully produced via combination of ring-opening reaction and condensation reaction. The results of FTIR and ^1H NMR confirmed that the degradable and low surface energy polyurethane was successfully prepared. Effect of physical properties on G-g-PLLA-PDMS-*b*-PU with the PDMS content was well studied. It was found that as

the PDMS content increased from 0 to 20%, the contact angle of the polyurethane increased from 71° to 108°. At the same time, the surface free energy of the polyurethane decreased from 52.19 to 11.74 N/m². A static hydrolysis experiment indicated that the hydrolytic degradation decreased with the increase of PDMS content. In addition, tensile strengths decreased from 21 to 14 MPa, and extensibility increased from 515 to 658%. The simulated ocean hanging template experiment proved that the G-g-PLLA-PDMS-*b*-PU copolymer material has the better antifouling performance, which indicates that the functionalized PU had a potential as a coating material for a marine antifouling coating application due to providing outstanding antifouling capacity and excellent mechanical properties.

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